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Testing a Motivation-Based Intervention for Mathematics and English Language Achievement among Senior Secondary School Students: A Randomized Pretest–Posttest Study Protocol

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Abstract

Background: Motivation is consistently associated with academic achievement, but recent longitudinal and meta-analytic evidence shows that the relationship is reciprocal and heterogeneous rather than a simple one-way causal pathway. Nigerian secondary-school studies also report associations between motivation and performance in Mathematics and English Language, while intervention evidence remains less common. **Objective:** This paper specifies a rigorous, reproducible protocol for evaluating whether a structured motivation-based intervention improves Mathematics and English Language Continuous Assessment Test (CAT) performance among senior secondary students. **Methods:** The documented design is a quantitative pretest–posttest randomized control-group study. At Christ Anglican College, Gwagwalada, Abuja, Nigeria, the accessible population is 200 students across SS1 (50), SS2 (60) and SS3 (90). A proportionate sample of 100 students is selected—SS1 = 25, SS2 = 30 and SS3 = 45—and randomly assigned to an experimental group ($n = 50$) and a control group ($n = 50$), with class-level representation preserved. The experimental condition is defined around goal setting, constructive feedback, encouragement, self-efficacy support, progress monitoring, recognition of improvement, study planning and persistence; the control condition continues normal academic activities. Mathematics and English Language achievement are measured at baseline and after the intervention using comparable school assessment procedures. **Planned analysis:** Subject-specific ANCOVA models estimate the intervention effect on post-intervention achievement while adjusting for corresponding pretest scores; descriptive statistics, baseline balance, change scores, effect sizes, confidence intervals, assumption checks and sensitivity analyses are also specified. **Contribution:** The protocol converts a broad motivation–achievement question into a falsifiable treatment-effect design, separates measured motivation from the manipulated intervention, and provides a transparent framework for reporting school-based experimental evidence without overstating causal claims before verified outcome data exist.

Keywords: *academic motivation; self-efficacy; Mathematics achievement; English Language achievement; randomized experiment; ANCOVA; secondary education; Nigeria*

1. Introduction

Academic achievement in secondary school is shaped by prior knowledge, instructional quality, opportunity to learn, home and school environments, self-beliefs, interests, goals, effort and persistence. Motivation is therefore important, but it should not be treated as a single trait or as an automatic explanation for low or high scores. Contemporary motivation theory distinguishes the reasons students engage, the value they attach to a

task, their expectancy of success, their sense of competence and agency, and the social conditions that sustain or undermine effort (Ryan & Deci, 2020; Eccles & Wigfield, 2020).

Recent evidence reinforces the need for careful causal language. Vu et al. (2024), synthesising 47 longitudinal studies in primary and secondary education, found a positive but modest average effect of prior motivation on subsequent achievement and a stronger reverse pathway from achievement to later motivation. Their findings imply that successful performance can itself strengthen motivational beliefs. A 2026 meta-analytic structural equation model likewise found interconnected relationships among intrinsic motivation, extrinsic motivation, self-efficacy and academic performance, with self-efficacy functioning as an important explanatory pathway (Shen et al., 2026). The practical implication is that an intervention must specify which motivational mechanism it intends to change and must measure academic outcomes independently rather than assume improvement.

Nigerian evidence points in the same broad direction while also revealing an evidence gap. Hassan, Gabriel and Hayatu (2025) reported an association between motivation and achievement in English Language and Mathematics among senior secondary students in Nasarawa State. Ekemezie and Okigbo (2026) found academic motivation to be among the predictors of Mathematics achievement in Anambra State, while Akeredolu and Olusakin (2026) reported a significant relationship between achievement motivation and English Language performance among senior secondary students in Lagos State. These studies are useful for contextual relevance, but correlational designs cannot determine whether deliberately strengthening motivation causes later achievement to improve. Recent Nigerian intervention studies in other school subjects show that experimentally structured learning conditions can alter both motivation and performance, but intervention effects may depend on context, learner characteristics and implementation fidelity (Nyikun & Musa, 2026; Osuh et al., 2026).

The present paper frames the motivation–achievement question as an intervention problem. It specifies a randomized pretest-posttest design in which a defined motivation-support package is compared with normal academic activity and Mathematics and English Language CAT outcomes are analysed after adjustment for baseline achievement. As a prospective protocol and analytical framework, the article reports no treatment-effect estimate before verified trial data are available. This preserves the planned design, prevents retrospective construction of findings and establishes an auditable analysis pathway for the completed study.

2. Theoretical and Empirical Foundation

2.1 Self-determination, competence and internalisation

Self-determination theory distinguishes intrinsic motivation from different forms of extrinsic motivation and proposes that more autonomous engagement is supported when learners experience competence, autonomy and relatedness (Ryan & Deci, 2020). In a school intervention, this means that motivation should not be reduced to praise or reward. Feedback must help students understand progress, goals must be achievable but meaningful, and teacher–student interaction should support agency rather than create dependence on external approval.

Competence is especially relevant to the present design because Mathematics and English Language provide repeated performance feedback through classwork, tests and continuous assessment. Students who experience repeated failure may lower expectations, disengage or avoid difficult tasks; conversely, visible progress can strengthen self-efficacy and persistence. The 2026 meta-analysis by Shen et al. supports the importance of self-efficacy as a pathway connecting motivation and performance.

2.2 Situated expectancy–value perspective

Situated expectancy–value theory explains achievement behaviour through students' expectations that they can succeed and the value they attach to the activity, including intrinsic value, utility value, attainment value and perceived cost (Eccles & Wigfield, 2020). For Mathematics and English Language, an intervention can therefore target both expectancy and value: students need credible evidence that improvement is possible and meaningful reasons for sustained engagement with the subject.

This perspective also discourages one-size-fits-all intervention claims. The same strategy may be motivating for one learner and ineffective for another because perceived difficulty, social expectations, prior performance and future goals differ. Held and Hascher (2023) demonstrated differential effects of a Mathematics motivation intervention across distinct motivation profiles, strengthening the case for documenting baseline heterogeneity and reporting subgroup analyses cautiously.

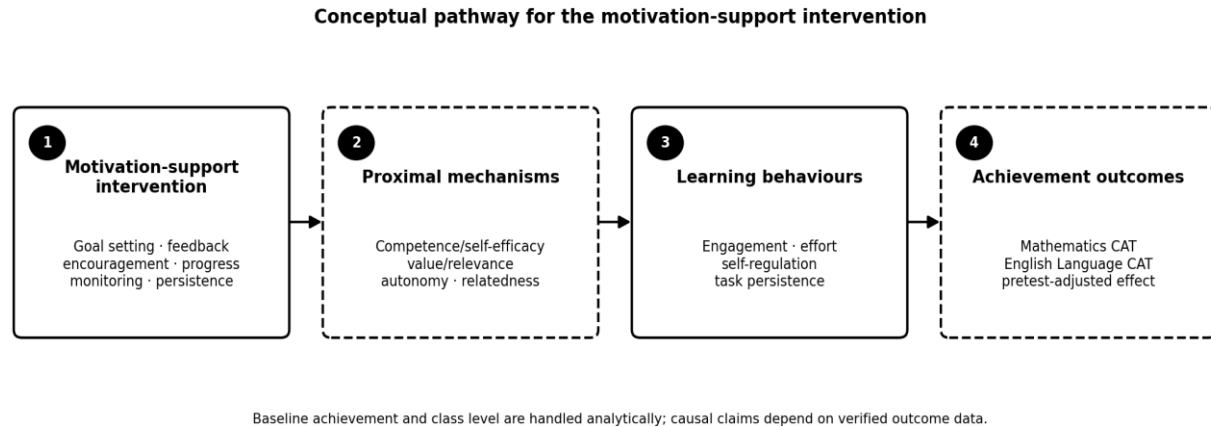


Figure 1. Conceptual pathway linking the motivation-support intervention to Mathematics and English Language achievement. Original framework for e059.

2.3 Why an experiment is needed in the Nigerian secondary-school context

Associations between motivation and achievement are educationally meaningful, but they are vulnerable to reverse causation and confounding. High-achieving students often report stronger self-beliefs because they have accumulated successful experiences; supportive teachers can simultaneously raise motivation and improve instruction; and family resources, prior attainment and school quality can affect both variables. Random assignment is therefore valuable because, when properly implemented, it reduces systematic baseline differences between treatment conditions and strengthens the interpretation of post-intervention differences.

Experimental evidence also demands restraint. Osuh et al. (2026) showed that a concrete extrinsic-reward intervention could influence academic performance in Nigerian secondary-school settings, whereas Nyikun and Musa (2026) reported improved motivation and Physics performance under a game-based strategy. These studies demonstrate that motivational conditions can be manipulated, but their mechanisms and transferability to routine Mathematics and English assessment are not guaranteed. The present protocol is designed to estimate the effect of a defined school-based motivation package under ordinary academic conditions.

3. Purpose, Research Questions and Hypotheses

The primary purpose is to determine whether exposure to a structured motivation-based intervention improves students' achievement in Mathematics and English Language after accounting for baseline achievement. A secondary purpose is to examine the relationship between measured motivation and achievement without confusing that association with the causal intervention effect.

3.1 Research questions

1. What are the baseline and post-intervention distributions of Mathematics and English Language achievement in the experimental and control groups?
2. After adjustment for Mathematics pretest performance, is experimental-group Mathematics achievement different from control-group achievement?
3. After adjustment for English Language pretest performance, is experimental-group English Language achievement different from control-group achievement?
4. How do Mathematics and English Language change scores compare between the two groups?
5. What relationship exists between baseline Student Academic Motivation Questionnaire (SAMQ) scores and Mathematics and English Language achievement, and how is that association distinguished from the randomized intervention effect?

3.2 Null hypotheses

H01: There is no statistically significant difference in post-intervention Mathematics achievement between the experimental and control groups after controlling for Mathematics pretest achievement.

H02: There is no statistically significant difference in post-intervention English Language achievement between the experimental and control groups after controlling for English Language pretest achievement.

H03: There is no statistically significant difference in Mathematics change scores between the experimental and control groups.

H04: There is no statistically significant difference in English Language change scores between the experimental and control groups.

H05: There is no statistically significant relationship between baseline SAMQ scores and Mathematics achievement.

H06: There is no statistically significant relationship between baseline SAMQ scores and English Language achievement.

4. Methods

4.1 Design

The study uses a quantitative experimental design, specifically a two-group pretest–posttest randomized control-group design. Selected students are randomly allocated to an experimental condition that receives the motivation-support intervention or to a control condition that continues normal instructional and academic activities. Achievement is measured before and after the intervention in Mathematics and English Language. The design is represented as R O1 X O2 for the experimental group and R O3 – O4 for the control group, where R denotes random assignment, O denotes measurement and X denotes the intervention.

4.2 Study setting and reporting boundary

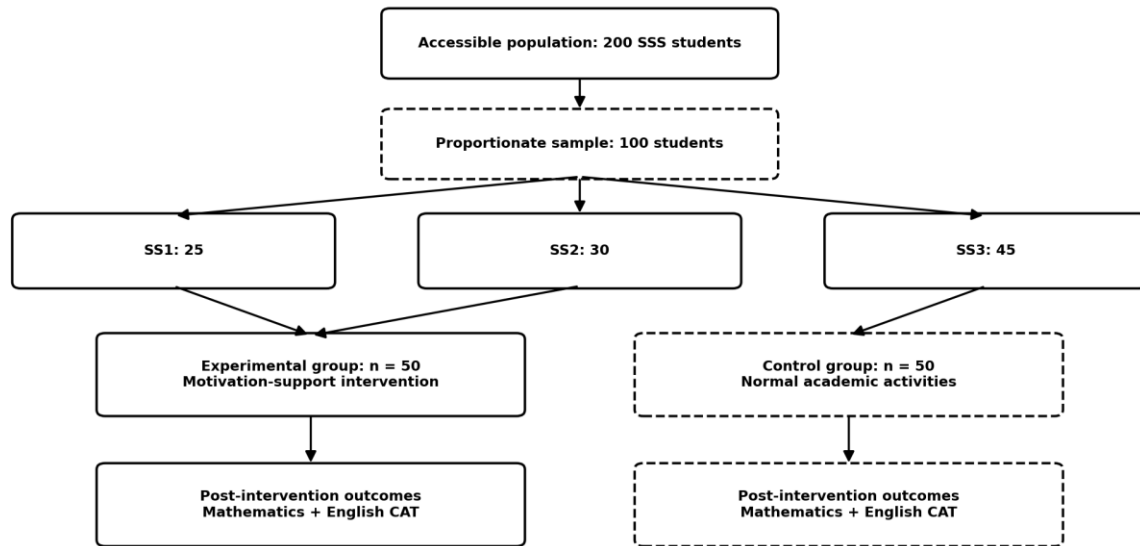
The study is set at Christ Anglican College, Gwagwalada, Abuja, Nigeria, during the 2025/2026 academic session. The school provides the authentic senior-secondary environment in which the SS1, SS2 and SS3 population, Mathematics and English Language instruction, motivation intervention and Continuous Assessment Test procedures are situated. Recruitment, authorization, consent/assent and intervention dates are recorded contemporaneously in the study file. The randomized designation applies only where individual allocation is implemented as specified; any class-level allocation is reported and analysed as a cluster-based or quasi-experimental design.

4.3 Population, sample and allocation

The accessible population at Christ Anglican College consists of 200 Senior Secondary School students: 50 in SS1, 60 in SS2 and 90 in SS3. A total sample of 100 students is selected using proportionate allocation, equivalent to 50% of each class-level population. The resulting sample is SS1 = 25, SS2 = 30 and SS3 = 45. After selection, participants are randomized to an experimental group ($n = 50$) and a control group ($n = 50$), with class-level representation preserved as closely as possible.

		Protocol sample	Sampling fraction
SS1	50	25	50%
SS2	60	30	50%
SS3	90	45	50%
Total	200	100	50%

Table 1. Population and proportionate sample specified by the study protocol.



Primary analysis: ANCOVA of post-intervention score adjusted for subject-specific pretest and class level.

Figure 2. Randomized study flow and primary analysis. Counts show the protocol-defined enrolment and allocation; the completed report presents the observed participant flow.

4.4 Sampling and random assignment

The sampling procedure is multistage. First, the population is stratified by SS1, SS2 and SS3. Second, the required number of students is selected at random within each class stratum. Third, selected participants are randomly assigned to the two study conditions. Randomization should be implemented with an auditable procedure such as a computer-generated random sequence and should be concealed from those enrolling participants until assignment is fixed.

The allocation target is SS1: 12 experimental/13 control; SS2: 15/15; SS3: 23/22, giving 50 students per condition. Because SS1 and SS3 contain odd sample counts, exact 1:1 balance is not possible within every class. Overall balance and class-level representation are therefore prioritised. The completed report presents the realized allocation and the auditable randomization method used.

4.5 Experimental condition: motivation-support package

The intervention comprises goal setting, constructive academic feedback, encouragement, self-efficacy support, progress monitoring, recognition of improvement, study planning and persistence strategies. Dose is treated as an implementation variable and is recorded contemporaneously through session date, duration, facilitator, attendance and component-delivery logs. Treatment-effect estimates are interpreted together with these fidelity records.

Minimum fidelity evidence		
Academic goal setting	Clarifies attainable targets and connects effort to observable progress.	Written student goal; review date; evidence of goal revision.
Constructive feedback	Strengthens competence beliefs by showing what was done well and what can be improved.	Feedback record tied to a real task or assessment.
Encouragement and self-efficacy support	Builds credible belief that improvement is possible through strategy and effort.	Teacher/research-assistant checklist; student reflection.
Progress monitoring	Makes improvement visible and supports self-regulation.	Simple progress record using classwork, practice or CAT-relevant indicators.
Recognition of improvement	Reinforces mastery and persistence without making rewards the sole reason for engagement.	Documented criterion for recognition; non-punitive application.
Study-goal planning and persistence	Translates motivation into repeatable behaviours and recovery after difficulty.	Short study plan; follow-up check; recorded barrier and response.

Minimum fidelity evidence		
Constructive teacher-student interaction	Supports relatedness, psychological safety and help-seeking.	Delivery log confirming respectful, non-coercive implementation.

Table 2. Intervention components and required fidelity evidence.

4.6 Control condition

Students in the control group continue with the school's normal academic activities without the structured motivation-support package during the intervention period. The final report should describe ordinary instructional exposure sufficiently to determine whether treatment contamination occurred. Where appropriate and ethically approved, the motivational programme may be offered after outcome data have been collected, but a wait-list arrangement should be reported only if it actually occurred.

4.7 Outcome and explanatory measures

Operational definition in the protocol		
Treatment assignment	Primary independent variable	Experimental motivation-support package versus control/normal activities.
Mathematics achievement	Primary outcome	Mathematics pretest and post-intervention/CAT score on comparable scoring scales.
English Language achievement	Primary outcome	English Language pretest and post-intervention/CAT score on comparable scoring scales.
Measured motivation	Secondary explanatory variable	Baseline score on the validated Student Academic Motivation Questionnaire (SAMQ).
Class level	Stratification/covariate	SS1, SS2 or SS3.
Baseline achievement	Covariate	Subject-specific pretest score used to adjust post-intervention comparisons.

Table 3. Variable roles and operational definitions.

4.8 Instrument quality

The motivation measure undergoes content review by qualified experts and empirical reliability assessment before inferential use. Where items form a common scale, internal consistency is estimated with Cronbach's alpha and, where feasible, coefficient omega. Reliability is calculated from the pilot or study data and reported as an observed statistic rather than assigned in advance.

Mathematics and English Language achievement measures are aligned in content domain, scoring procedure and difficulty across baseline and post-intervention assessment to the extent feasible. For school CATs, the study record documents item count, maximum score, duration, marking procedure, administration date and whether identical or parallel forms are used. Where appropriate for dichotomously scored items, KR-20 or another justified reliability estimate is reported.

4.9 Ethical safeguards

School-based experiments involving adolescents require clear administrative authorization, age-appropriate information, applicable parent/guardian consent and student assent procedures, protection from academic penalty for non-participation, secure handling of identifiable records and a plan for responding to distress or perceived coercion. Routine CAT administration is not equivalent to covert research participation; the use of assessment records for research must remain within the approved ethical and administrative framework.

Implementation proceeds only under documented school authorization and the applicable consent/assent requirements for research involving secondary-school students. Approval and authorization records are retained in the study file and reported in the completed trial report.

5. Statistical Analysis Plan

5.1 Data preparation and descriptive analysis

Each participant should receive an anonymous study identifier. Data checks should identify duplicate records, impossible scores, missing observations, allocation inconsistencies and values outside instrument ranges. Cleaning decisions should be logged rather than silently applied. Baseline and post-intervention achievement

should be summarized by group using n , mean, standard deviation, minimum, maximum and 95% confidence intervals where appropriate.

Baseline comparability should be described rather than judged only by significance tests. Randomization can produce chance imbalances, especially in a sample of 100. Differences in subject pretests and class-level composition should therefore be presented directly. The primary treatment-effect model adjusts for the corresponding baseline score regardless of whether the baseline group difference is statistically significant.

5.2 Primary ANCOVA models

ANCOVA is the primary analysis because the design includes baseline and post-intervention measurements. Adjusting the post-intervention score for the corresponding pretest generally provides a clear treatment-effect estimate and avoids some limitations of analysing posttest scores or change scores alone (Vickers & Altman, 2001).

$$\text{MathPost}_i = \beta_0 + \beta_1(\text{Treatment}_i) + \beta_2(\text{MathPre}_i) + \beta_3(\text{Class}_i) + \varepsilon_i$$

$$\text{EnglishPost}_i = \beta_0 + \beta_1(\text{Treatment}_i) + \beta_2(\text{EnglishPre}_i) + \beta_3(\text{Class}_i) + \varepsilon_i$$

In each model, β_1 is the adjusted experimental-versus-control difference. The principal report should include adjusted group means, the treatment coefficient or equivalent F test, degrees of freedom, p value, 95% confidence interval and an effect-size measure such as partial η^2 or a standardized adjusted mean difference. Statistical significance is evaluated at $\alpha = .05$, but interpretation should emphasize magnitude and uncertainty rather than a binary threshold.

5.3 Supplementary analyses

Change scores (posttest minus pretest) may be summarized and compared as a secondary analysis. Baseline SAMQ scores are examined with correlation and regression analyses to estimate their association with achievement. These analyses answer a different question from the randomized treatment effect: they describe the relationship between a measured psychological variable and performance rather than the effect of assignment to the intervention.

Exploratory interactions by class level may be considered if cell sizes and model stability permit, but the study is not powered to make strong subgroup claims. Any subgroup analysis should therefore be labelled exploratory and interpreted with confidence intervals. Because motivation interventions can affect students differently depending on baseline profiles (Held & Hascher, 2023), reporting heterogeneity is useful, but small-sample overinterpretation should be avoided.

5.4 Assumptions, missing data and sensitivity

Issue		
Linearity	Plot residuals/relationship between pretest and posttest.	Use transformations or robust approaches only when justified and report the decision.
Homogeneity of regression slopes	Test treatment $\times$ pretest interaction.	If strongly violated, report an interaction model rather than a single common slope.
Residual distribution/influence	Residual plots and influence diagnostics.	Investigate data errors; use robust inference or sensitivity analysis for legitimate influential cases.
Missing outcome data	Report count and reason by group.	Do not silently drop cases; describe complete-case and, if appropriate, principled missing-data sensitivity analyses.
Treatment contamination	Implementation/fidelity records.	Describe exposure in both groups and interpret effect estimates in light of contamination.
Protocol deviations	Compare planned and actual procedures.	List deviations before interpreting outcomes; do not rewrite the protocol retrospectively.

Table 4. Analysis and validity controls.

6. Intervention Fidelity, Transparency and Reporting

A treatment effect is interpretable only when the intervention itself is described well enough to reproduce. The TiDiER framework emphasizes the need to report what was delivered, by whom, where, when, how much, and whether delivery was modified or monitored (Hoffmann et al., 2014). For this protocol, fidelity records

should therefore include attendance, session date, facilitator, component delivered, duration, deviations, participant engagement and any adverse or unintended response.

Transparent randomized-trial reporting is also essential. CONSORT 2025 calls for explicit reporting of trial design, eligibility, allocation, interventions, outcomes, participant flow, analysis and deviations rather than leaving readers to infer what probably happened (Hopewell et al., 2025). Although this educational study is not a drug trial, the underlying reporting principles—prospective specification, participant-flow accounting, clear treatment descriptions and complete outcome reporting—are directly valuable for school-based experiments.

The completed trial report presents participant flow for numbers assessed, randomized, receiving the assigned condition, completing each assessment and included in each analysis. All prespecified primary outcomes are reported regardless of statistical significance, and any deviation from this protocol is identified and justified.

7. Expected Contribution to Educational Practice

This protocol is designed to move practice beyond generic advice to “motivate students.” It defines motivation support as a set of observable, teachable practices linked to theory: clear goals, competence-supportive feedback, relevance, progress monitoring, recognition of improvement, planning, persistence and constructive relationships. If the experimental group later demonstrates an advantage after baseline adjustment, the result would support the tested package under the specific implementation conditions—not the broader claim that motivation alone determines achievement.

For teachers, the intervention can be integrated with ordinary instruction rather than treated as a separate counselling programme. Goal setting and feedback can be attached to real Mathematics problem solving and English reading/writing tasks. Progress monitoring can use authentic class evidence. For school leaders, the protocol provides a basis for evaluating a low-cost academic-support strategy with explicit fidelity measures rather than relying on testimonials or uncontrolled before–after comparisons.

For research, the design addresses a gap between correlational Nigerian evidence and causal intervention testing. Recent local studies demonstrate that motivation is associated with Mathematics and English performance (Hassan et al., 2025; Ekemezie & Okigbo, 2026; Akeredolu & Olusakin, 2026), while experimental studies show that motivational conditions can be changed in school contexts (Nyikun & Musa, 2026; Osuh et al., 2026). A well-reported randomized study can test whether a deliberately structured motivation-support package adds value beyond normal school activity.

8. Limitations and Boundaries

The protocol concerns one school population of 200 students with a planned sample of 100, so external validity will be limited even if internal validity is strong. Mathematics and English Language outcomes may respond differently to the same intervention because subject demands, prior confidence and teacher practices differ. The sample also limits the precision of subgroup analyses.

The central limitation is intrinsic to a protocol article: no treatment-effect estimate is presented before completion and verification of the trial dataset. The study is confined to Christ Anglican College, Gwagwalada, Abuja, Nigeria, and any completed results must therefore be interpreted within that school context. Empirical conclusions depend on verified implementation, participant-flow, instrument-quality and reproducible statistical records.

Finally, measured motivation and the randomized treatment are conceptually distinct. A motivation questionnaire may predict achievement without proving that changing the questionnaire score causes achievement to change. Conversely, an intervention may improve performance through mechanisms not fully captured by a single motivation scale. The analysis should therefore keep treatment assignment, measured motivation, engagement and achievement analytically separate.

9. Conclusion

The e059 protocol provides a defensible framework for testing a motivation-based intervention in senior secondary education without converting plausible theory into unverified findings. It specifies a population of 200 students, a proportionate sample of 100, 50/50 randomized allocation, Mathematics and English Language

pretest-posttest outcomes and ANCOVA-based primary analysis. Contemporary motivation theory, explicit fidelity controls, ethical safeguards and transparent reporting principles strengthen the design. The protocol therefore establishes a reproducible basis for estimating the intervention effect once the study is implemented and the resulting data are verified and analysed as specified.

Declarations

Ethics and participant protection

The protocol requires documented school authorization and applicable parent/guardian consent and student assent safeguards before recruitment and use of student assessment records for research.

Data availability

No participant-level outcome data are reported in this protocol article. The completed study report will describe the de-identified analytical dataset, access conditions and restrictions consistent with consent and institutional policy.

Funding

No external grant funding is asserted in this protocol.

Competing interests

Competing-interest disclosure is managed through the journal's signed author-declaration process and retained with the editorial record for the Version of Record.

Author contribution and accountability

Tanimu Zachariah is the sole named author of e059 and accepts responsibility for the study protocol, implementation records, ethical documentation, dataset, analysis and final manuscript. The study setting is Christ Anglican College, Gwagwalada, Abuja, Nigeria.

Generative AI and editorial support

Editorial and document-production assistance was used to improve structure, methodological transparency and presentation. Such assistance is not an author or source of evidence. The named author and journal editors remain responsible for every factual claim, citation, methodological statement and publication decision.

Figure and table provenance

Figures 1–2 and Tables 1–4 were created specifically for e059 to organize the study design and analytical framework. They contain no participant-level results or third-party images.

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